

Measurement of selected preventable adverse drug events in electronic health records: Toward developing a complexity score

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Purpose. The defining of a select number of high-priority preventable adverse drug events (pADEs) for measurement in the electronic health record (EHR) and the estimation of pADE incidences in two tertiary care facilities are described.

Methods. This study was part of a larger effort aimed at developing an automated electronic health record (EHR)-based complexity-score (C-score) that ranks hospitalized patients according to their risk for pADEs for clinical intervention. We developed measures for 16 high-priority pADEs often deemed preventable using discrete clinical and administrative EHR data. For each pADE we specified inclusion and exclusion criteria that were used to define risk populations for each specific pADE. The incidence of each type of pADE was then measured during a designated follow-up period considering all adult admissions to 2 large academic tertiary care hospitals, who were eligible for the pADE-specific risk populations during any of their first 5 hospital days.

Results. Utilizing the data from 83,787 admissions who were at risk for at least one pADE during at least one of their first five hospital days, we found that 27,193 admissions (32.5%) developed at least one pADE. Uncontrolled postsurgical pain, uncontrolled pneumonia, and drug-associated hypotension had the highest incidences with the following number of days with pADE per number of patients at risk: 13,484 of 19,640; 527 of 1,530; and 13,394 of 43,630, while drug-associated falls (446 of 75,036), drug-associated acute mental status changes (262 of 66,875) and venous thromboembolism (214 of 74,283) had the lowest incidence rates.

Conclusion. EHR-based definitions of clinically important pADEs were developed, and the incidence of the pADEs was estimated. These definitions will be advanced for the creation of prediction models to develop a C-score for identifying patients at risk for pADEs to prioritize pharmacist intervention.

Keywords: adverse drug event, C-score; electronic health records; incidence; inpatient; medication errors, prevention and control

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An adverse drug event (ADE) is defined as an adverse outcome or a patient injury such as physical or mental harm or loss of function resulting from a medical intervention related to a drug.¹ ADEs are common during hospitalization. According to previous observational studies of ADEs in hospital settings,

the frequency of ADEs ranges from 2.4 to 16.2 per 100 hospital admissions, and about 25–36% of ADEs are judged to be preventable.^{2–8} Beyond clinical harm, ADEs may escalate hospital costs, with estimates of \$1.6 billion–\$5.6 billion total spent per year in hospitals nationwide to treat patients who suffer ADEs.⁹

To monitor trends in ADE rates, understand the etiology of ADEs, and set targets for patient safety programs, it is important to track ADE rates over time and across hospital systems. Approaches to measure ADE occurrences include spontaneous reports, systematic expert chart reviews, and computer-based surveillance systems.¹⁰ While chart review has been used widely in previous studies involving ADE detection, it remains an impractical means for routine ADE identification due to the high cost and resource burden associated with the manual review process.^{11,12} In the United States, mandatory and voluntary reporting programs exist to measure medication errors and preventable ADEs (pADEs) in order to support quality improvement.¹³ However, error reporting depends on individuals' ability to identify and willingness to report such incidents, which results in imprecise measurement of pADE frequency.^{10,11,14,15} While computer-based surveillance systems using explicit definitions that pinpoint discrete information in electronic health records (EHRs) have demonstrated better sensitivity for identifying pADEs,^{16,17} the numbers of validated measures and the breadth of pADEs that can be tracked are limited. Furthermore, previous studies that used computer-based surveillance systems to track ADEs heavily relied on simplified signals (e.g., a single abnormal laboratory result or an elevated drug concentration) using nonintegrated information, which limits the specificity of these measures and often requires manual verification.¹⁷⁻¹⁹ An enhanced approach that accurately captures clinically significant ADEs using automated and practical algorithms is needed.

A comprehensive understanding of patient characteristics and care processes associated with pADEs will help guide preventive interventions and improve patient safety. Perhaps due to the difficulty in measuring pADEs, research characterizing patient populations at high risk for pADEs is limited.^{20,21} As part of its Pharmacy Practice Model Initiative, now termed the Practice Advancement Initiative, to

KEY POINTS

- Using discrete fields from fully integrated electronic health records, this study developed measures for 16 types of preventable adverse drug events (pADEs) in hospitals.
- Nearly one third of admissions were associated with at least 1 pADE during hospitalization.
- Uncontrolled postsurgical pain, drug-associated hypotension, and uncontrolled hypertension had the highest incidence in patients at risk, while uncontrolled *Clostridium difficile* infections, venous thromboembolic events, and drug-associated mental status changes were least common.

optimize medication-use outcomes,²² the ASHP Research and Education Foundation sponsored a study to develop a complexity score (C-score) that would use automated EHR data in prospectively identifying hospitalized patients at high risk for pADEs.²³ The C-score is designed to rank hospitalized patients for prioritization of drug therapy management interventions according to their aggregate risk for clinically important pADEs. This report is the second of 3 articles on C-score development. The first article described a comprehensive literature review and a stakeholder survey used to select clinically important pADEs for inclusion in the C-score.²⁴ The current article focuses on the development of operational definitions for the selected pADEs and estimation of their incidence in hospitalized patients using discrete clinical and administrative data obtained from EHRs. The third and final article will describe prediction models for each of the pADEs based on definitions provided in the current article.

Methods

Selection of pADEs. The methods used to select pADEs have been described previously.²⁴ In brief, we conducted a comprehensive literature review to identify studies describing the overall epidemiology and nature of pADEs acquired during hospitalization. Extracted concepts were summarized in a survey sent to ASHP members and presented to a national technical expert panel (TEP) that was assembled to support this study. A total of 21 pADEs evolved from this process. Several pADEs were discarded after further assessments due to (1) their inability to be measured in EHRs (i.e., drug-induced seizures and new-onset stress ulcers), (2) too few outcomes to allow prediction modeling (i.e., drug-induced respiratory depression, drug-induced hemorrhage, and drug-induced ileus), or (3) their unsuitability for pharmacist intervention or limited preventability (i.e., manifestation of hospital-acquired infection and drug-induced blood dyscrasia). We developed separate measures for certain types of electrolyte imbalances (hypokalemia, hyperkalemia, and hyponatremia) and uncontrolled infections (uncontrolled *Clostridium difficile* infection and uncontrolled pneumonia), resulting in a total of 16 pADEs that were targeted for measurement.

Study population. The patient population for pADE definition development included adult patients (≥18 years old) admitted to UF Health Shands from January 1, 2012, through October 31, 2013, or UF Health Jacksonville from March 1 through October 31, 2013. The shorter time frame for admissions to UF Health Jacksonville was chosen due to later implementation of the current fully integrated EHR system (Epic Systems Software, Epic, Verona, WI). We excluded patients discharged on the same day of admission but imposed no other restrictions in order to focus C-score development on the entire population typically encountered in a hospital (e.g., including patients admitted for observation or

delivery). We conducted a descriptive analysis of demographic (i.e., age, sex, ethnicity, and race) and clinical (i.e., the primary reason for hospitalization, length of stay, and patient type) characteristics in the study population. Using the Clinical Classifications Software developed as part of the Healthcare Cost and Utilization Project,²⁵ we categorized patients' principal diagnoses and present the most frequent categories.

pADE definitions. We aimed to develop pADE definitions using discrete data elements in the hospitals' EHR system, such as laboratory results, vital signs, patient assessment results (e.g., stool abnormality), physician orders for diagnostic tests, pathogen culture results, and initiation of drugs treating adverse events. Composites of these elements were considered to improve the specificity of several pADE definitions.

Development of definitions for each of the 16 pADEs was supported by 3 main sources: literature review, TEP discussions, and iterative refinements based on empirical analysis in our testing population. First, we reviewed the definitions of adverse events that had been used in landmark studies or clinical guidelines. These definitions were then presented to the TEP, which was composed of experts in medication safety and clinical pharmacy, to select ADEs that were (1) considered clinically significant, defined as either resulting in patient harm or having the potential to cause harm based on published evidence, (2) deemed preventable based on evidence, and (3) have a frequency of at least 1 event per 1,000 admissions to allow prediction modeling for C-score development. For example, in selecting a threshold that would define drug-associated hypoglycemia, we considered serum or capillary blood glucose concentrations between 70 and 40 mg/dL. The higher threshold was prevalent but not likely to cause harm and not a good focus for preventive action. The lower threshold was considered a catastrophic event that

should never happen and a good focus for prevention but had too small a frequency and thus restricted the development of stable prediction models. The final definition of each pADE was informed by exploratory analyses that considered the frequency and validity of the various options and the potential biases due to data availability or data reliability. The relative importance of each of these 3 steps differed for each pADE. For pADEs with widely accepted diagnostic criteria and validated definitions and measurement (e.g., venous thromboembolic events), panel review or refinement processes were less involved. However, for pADEs without validated measurement or a range of definitions or with anticipated problems in measurement, we relied heavily on panel review and exploratory analyses to develop our case-identification algorithms and confirmed the algorithm via expert chart review (e.g., uncontrolled pneumonia, uncontrolled *C. difficile* infection). All expert chart reviews were conducted by a clinical pharmacist specialist who specialized in health information technology and were focused on verifying the accuracy of our measures in identifying the manifestation of an adverse event and not to assess preventability.

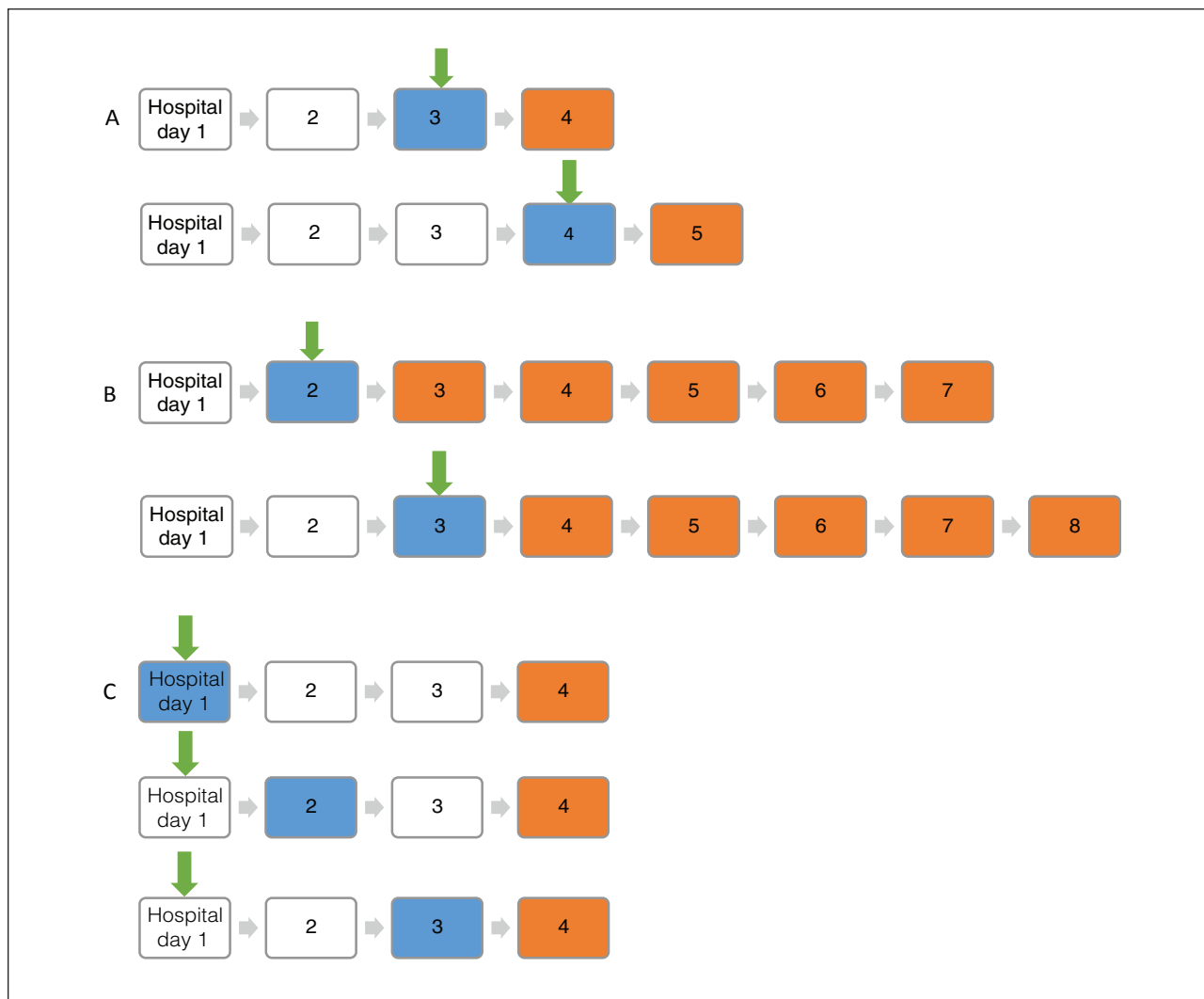
Establishment of pADE cohorts.

To develop measures that could estimate pADE incidence, we needed to define, in addition to the adverse event itself, the risk population for each pADE (i.e., inclusion and exclusion criteria) and a window during which hospitalized patients are at risk for the pADE. The 16 pADEs were conceptualized as ADEs that result from an error of commission (a contraindicated therapy or medication overdose caused an adverse event) or an error of omission (lack of or insufficient drug therapy caused an adverse event). For pADEs that are generally caused by an error of commission (i.e., drug-associated acute mental status changes, acute kidney injury [AKI], acute liver injury [ALI], falls, hypoglycemia, hypotension, and QT-interval prolon-

gation), we required an exposure to a risk drug before the pADE. In other words, we determined a pADE occurrence only after an offending drug was given in order to focus our adverse event measures on causal pathways that involve inappropriate drug therapy and reduce alternative explanations for the adverse event. We defined a risk day as the day when the risk drug was given. In pADE ascertainment, we followed patients until a certain timepoint after a risk day (Figure 1). The length of follow-up differed among the 16 pADEs depending on plausible proximity between a risk drug and the adverse event. For example, for ascertainment of drug-associated hypotension, we required exposure to hypotensive agents and followed patients until the following day to determine whether they experienced severe hypotension. In contrast, since nephrotoxicity may develop over several days, we followed patients for up to 5 days after a risk day, which was defined by exposure to nephrotoxic drugs. Importantly, no further causality assessments were performed and the etiology of the adverse event could still be multifactorial. Because patients may be exposed to risk drugs over several days, patients could have several risk days, each followed by the same specified time period to ascertain the pADE.

Among pADEs resulting from potential errors of omission, uncontrolled *C. difficile* infection, pneumonia, and postsurgical pain, patients were assessed once they became at risk for the adverse events (i.e., once *C. difficile* infection manifested, once patients were admitted for pneumonia, or after a surgical procedure). To ascertain uncontrolled *C. difficile* infection, we looked for persistent clinical symptoms (e.g., diarrhea) after 10 days from the first risk day, which was the onset of *C. difficile* infection. For uncontrolled pneumonia, we assessed failure to achieve respiratory stability and control of clinical symptoms (i.e., fever, tachycardia or leukocytosis) at the fourth day of hospitalization

Figure 1. Risk days and follow-up days for a variety of preventable adverse drug events. For drug-associated hypotension (panel A), risk days include hospital days 3 and 4 because of exposure to hypotensive drugs. Follow-up days, during which onset of hypotension is determined, include hospital days 4 (for risk day 3) and 5 (for risk day 4). For drug-associated acute kidney injury (panel B), follow-up days for risk day 2 include hospital days 3–7, and follow-up days for risk day 3 include hospital days 4–8; this is because acute kidney injury can develop over several days after exposure to nephrotoxic drugs. For uncontrolled pneumonia (panel C), the event is defined as failure to control symptoms at hospital day 4 among patients with pneumonia at admission, so hospital day 4 is the only follow-up day for risk day 1, 2 or 3. The green arrow represents exposure to the inciting agent or condition (a hypotensive drug in panel A, a nephrotoxic drug in panel B, or pneumonia on admission in panel C), the blue boxes represent risk days, and the orange boxes represent follow-up days.



among patients with pneumonia on admission. Uncontrolled postsurgical pain was determined for each of the first 5 days after surgery. We did not require any qualifying event or drug exposure in the assessment of hyperkalemia, hypokalemia, hyponatremia, uncontrolled hyperglycemia, uncontrolled hypertension, and

venous thromboembolism (VTE) ascertainment, and, accordingly, each of the first 5 hospital days was a risk day. The onset of pADEs was evaluated between the second hospital day and the sixth hospital day (i.e., on each day following a risk day), except for VTE. The onset of VTE was ascertained with the presence of *International Classifi-*

cation of Diseases, 9th Edition, Clinical Modification (ICD-9-CM) codes for VTE accompanied with an initiation of VTE treatment or diagnostic tests (i.e., ultrasound, chest computed tomography, and ventilation/perfusion scan) within 30 days of hospitalization.

We furthermore developed pADE specific exclusion criteria that re-

moved patients with a disease or a clinical symptom that could be a strong alternative cause for ADE development. For example, we excluded patients with dementia in the measure for drug-associated acute mental status changes and excluded patients with VTE present on admission in the measure for hospital-acquired VTE.

Incidence estimation. The incidence of pADEs—in terms of duration per admission in the risk population—was calculated retrospectively from EHR data extracted for the previously described study sample as the number of pADE events during the specified follow-up period divided by the total number of patients whose risk days coincided with at least 1 of their first 5 hospital days. Note that each pADE measure was defined for the purposes of C-score development, which was to develop prediction models for the first 5 hospital days. In accordance with the C-score, we limited the risk days to the first 5 hospital days; discharged patients were assumed to not have a pADE.

For most pADEs, we allowed recurrence and counted all days with pADE manifestation during follow-up if the pADE occurred during the defined risk windows after a risk day. Exceptions included drug-associated AKI, uncontrolled *C. difficile* infection, uncontrolled pneumonia, drug-associated ALI, drug-associated QT-interval prolongation, and VTE, because repeated occurrences during 1 admission are rare and because of difficulties in distinguishing the new onset of a subsequent event from residual signs and symptoms of the preceding event.

Measures development and calculation of incidence were conducted using SAS, version 9.4 (SAS Software, Cary, NC). This study was approved by the institutional review and privacy boards of the University of Florida.

Results

This study included 83,787 admissions—33,935 (40.50%) were patients older than 60 years and 46,218

(55.16%) were women—that contributed a total of 288,323 risk days with various follow-up periods thereafter. The mean patient age was 53.2 years. The mean length of hospital stay was 6 days, yet the majority of the study population stayed 5 or fewer days (74.4%). The top diagnoses resulting in hospital admission in both institutions involved the circulatory system, injury, and pregnancy (Table 1).

We defined 16 pADEs using discrete data from the clinical or administrative portion of the hospitals' EHRs. Data elements used to measure pADEs included vital signs and patient status assessments (e.g., pain scores, mental status change assessments), electronic medication administration records, the medication list before hospital admission, laboratory test values, electrocardiogram readings, diagnosis codes, procedures, and the hospitals' incident-reporting system. If diagnosis codes (*ICD-9-CM* codes) were used to identify the adverse event, we used clinical procedures or other interventions to time the onset of the event. For example, onset of VTE was defined based on related radiological tests or initiation of anticoagulation at treatment doses, but diagnosis codes were required to confirm occurrence of the event. For definitions that relied on laboratory test values, which are known to have some level of measurement error, we used algorithms that required a subsequent intervention (e.g., hypokalemia had to be followed by administration of potassium) or repeated measurements to confirm the validity of the original finding (e.g., hypoglycemia based on a blood glucose concentration of <50 mg/dL was only counted if it was not followed by a subsequent value of >80 mg/dL within 10 minutes). Measurements taken in the operating room were not considered. The hospitals' incident-reporting system was used to detect drug-associated falls. Table 2 shows the finalized definitions for each of the 16 pADEs as well as definitions for risk days and exclusion criteria, if applicable.

Among the 83,787 admissions included in this study, 27,224 (32.49%) developed at least one pADE (Table 3). More than 1 pADE occurred during 9.6% of admissions. The pADEs accounting for the largest number of events during pADE specific follow-up were uncontrolled postsurgical pain (13,484), drug-associated hypotension (13,394), uncontrolled hypertension (10,880), and hyperglycemia (9,819) (Table 4). Uncontrolled postsurgical pain (68.66%), uncontrolled pneumonia (34.44%), drug-associated hypotension (30.70%), uncontrolled *C. difficile* infection (26.07%) had the highest incidences when considering the actual risk population. VTE, drug-associated acute mental status changes, drug-associated falls, and drug-associated ALI the lowest frequency with less than 1 event per 100 at-risk admissions.

Discussion

This report summarizes the development of EHR-based measures of common pADEs and provides frequency estimates from 2 tertiary care hospitals. Several findings warrant further discussion.

First, taking advantage of the plethora of information in fully integrated EHRs, we were able to develop measures that can determine the occurrence and time of onset of pADEs. While a discussion of individual measures validity goes beyond the scope of this paper, we would like to highlight that several pADE measures used composites of diagnoses or laboratory findings with clinical action. This approach ensured pADE manifestation and clinical relevance, in contrast with computer-based pADE detection algorithms that rely on simplified signals (e.g., a single abnormal laboratory result or elevated drug concentration) using nonintegrated information. Such composites also allowed measurement of pADEs without designated diagnosis codes (e.g., uncontrolled infections, acute mental status changes) or where diagnosis codes are underutilized (e.g., drug-associated AKI, drug-associated ALI). The use of test-

Table 1. Demographic and Clinical Characteristics of the Study Population

Variable	UF Health Shands	UF Health Jacksonville	Total
No. unique patients	42,593	13,314	55,907
No. unique admissions	65,518	18,269	83,787
No. patient-days in analysis	231,703	56,620	288,323
Mean age, yr	53.8	51.2	53.2
Patient age, no. (%) patients			
18 or 19 yr	1,438 (2.19)	255 (1.40)	1,693 (2.02)
20–29 yr	8,524 (13.01)	2,672 (14.63)	11,196 (13.36)
30–39 yr	7,102 (10.84)	2,285 (12.51)	9,387 (11.20)
40–49 yr	8,113 (12.38)	2,710 (14.83)	10,823 (12.92)
50–59 yr	12,434 (18.98)	4,319 (23.64)	16,753 (19.99)
60–69 yr	13,066 (19.94)	3,210 (17.57)	16,276 (19.43)
70–79 yr	9,277 (14.16)	1,654 (9.05)	10,931 (13.05)
80–89 yr	4,595 (7.01)	932 (5.10)	5,527 (6.60)
≥90 yr	969 (1.48)	232 (1.27)	1,201 (1.43)
Female	35,933 (54.84)	10,285 (56.30)	46,218 (55.16)
Hispanic	2,360 (3.60)	665 (3.64)	3,025 (3.61)
Race			
White	46,338 (70.73)	8,018 (43.89)	54,356 (64.87)
Black or African American	14,041 (21.43)	9,093 (49.78)	23,134 (27.61)
Asian	419 (0.64)	161 (0.88)	580 (0.69)
Native Hawaiian or other Pacific Islander	360 (0.55)	17 (0.09)	377 (0.45)
American Indian or Alaska Native	91 (0.14)	22 (0.12)	113 (0.13)
Other	3,371 (5.15)	784 (4.29)	4,155 (4.96)
Unknown	898 (1.37)	174 (0.95)	1,072 (1.28)
Top 12 condition categories (based on principal diagnosis)			
Diseases of the circulatory system (chronic)	9,770 (14.92)	2,282 (12.51)	12,052 (14.38)
Injury and poisoning (nonchronic)	9,556 (14.59)	2,527 (13.85)	12,083 (14.42)
Complications of pregnancy, childbirth, and puerperium	5,539 (8.46)	2,118 (11.61)	7,657 (9.14)
Neoplasms (chronic)	4,432 (6.77)	571 (3.13)	5,003 (5.97)
Symptoms, signs, and ill-defined conditions (nonchronic)	4,406 (6.73)	1,998 (10.95)	6,404 (7.64)
Diseases of the digestive system (nonchronic)	4,307 (6.58)	858 (4.70)	5,165 (6.16)
Infectious and parasitic disease (non-chronic)	2,738 (4.18)	982 (5.38)	3,720 (4.44)
Diseases of the genitourinary system (nonchronic)	2,338 (3.57)	631 (3.46)	2,969 (3.54)
Diseases of the respiratory system (nonchronic)	2,312 (3.53)	...	2,312 (2.76)
Diseases of the musculoskeletal system (chronic)	2,116 (3.23)	...	2,116 (2.53)
Mental disorders (chronic)	...	1,239 (6.79)	1,239 (1.48)
Endocrine, nutritional, and metabolic diseases and immunity disorders (chronic)	...	598 (3.28)	598 (0.71)

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Table 1. Demographic and Clinical Characteristics of the Study Population

Variable	UF Health Shands	UF Health Jacksonville	Total
Mean length of stay, hr	139	137	139
Length of stay, no. (%) patients			
<24 hr	4,077 (6.22)	1,393 (7.62)	5,470 (6.53)
24 to <48 hr	12,763 (19.48)	3,914 (21.42)	16,677 (19.90)
48 to <72 hr	11,766 (17.96)	3,928 (21.50)	15,694 (18.73)
72 to <96 hr	8,756 (13.36)	2,261 (12.38)	11,017 (13.15)
96 to <120 hr	5,846 (8.92)	1,459 (7.99)	7,305 (8.72)
120 to <144 hr	4,370 (6.67)	970 (5.31)	5,340 (6.37)
≥144 hr	17,940 (27.38)	4,344 (23.78)	22,284 (26.60)
Patient type			
Inpatient	57,490 (87.75)	15,307 (83.79)	72,797 (86.88)
Observation	8,028 (12.25)	2,962 (16.21)	10,990 (13.12)

ing and service times allowed for the timing of pADE manifestation, which in turn established proximity to risk windows. To assess pADEs caused by errors of commission or omission, exact timing was critical to reduce other causal pathways of the event.

Second, the selected pADEs occurred frequently, with nearly one third of admissions with at least 1 pADE. Although previous studies have reported lower ADE frequencies (16–23%) than our study, those results need to be interpreted with caution.^{7,26–29} Frequency estimation of ADEs is known to be influenced by the chosen measurement methods (e.g., chart review, patient interviews, voluntary incident reports, billing data analysis, computerized surveillance system), study population, and criteria used for defining events. One study that compared computer-based algorithms, chart review, and stimulated reports found that computer-based algorithms detected more events than voluntary reports but less than chart review, which can consider a broader range of pADE types.¹⁰ Of note, due to our study's overarching focus to develop pADE prediction models, we

restricted frequency measures to the first 5 hospital days but allowed recurrence of some pADE types during the examined risk windows, while other studies may have not. Like most previous studies, we did not ascertain pADEs after discharge, which may underestimate frequency rates, especially for those pADEs with delayed onset, such as VTE.

Third, uncontrolled postsurgical pain had the highest absolute number of pADE days (13,484 days) and the greatest impact (68.66 pADE days per 100 admissions) in its risk population. While the impact was higher than that in any report that has comprehensively measured pADE frequency, it is important to note that measurement was confined to postsurgical patients. In fact, previous studies with specific focus on postsurgical populations have reported that 31.5–80.0% of patients experienced uncontrolled pain after surgery, which is comparable with our results.^{30–32} We chose the postsurgical setting to focus on acute pain management and minimize the contribution of more-complicated scenarios involving management of patients with chronic pain or opioid use dis-

order or both. Considering the prescription opioid epidemic, elevated pain scores could be interpreted differently in such scenarios and argue against their classification as a pADE. However, postsurgical management of acute pain should remain a high priority, especially because adverse acute sequelae of acute opioid use in hospitals (e.g., respiratory depression, ileus) are rare.^{33,34}

Drug-associated hypoglycemia occurred in 5.75% of patients receiving antidiabetic drugs in our study. While general frequency rates of hypoglycemia including drug-related and non-drug-related events are generally reported at lower frequencies (0.48–2.82%),^{6,7,35} frequencies of severe hypoglycemia have been estimated at 8–10% in studies that focused on inpatients who received antidiabetic medications, as was done in our investigation.^{36–38}

Consistent with other reports, VTE, drug-associated acute mental status changes, drug-associated falls, and drug-associated ALI occurred the least often, with frequencies of fewer than 1 pADE day per 100 at-risk admissions. While frequencies may sug-

Table 2. Definitions of pADEs and Associated Risk Days^a

pADE	Risk Day Definition (Cohort Entry)	pADE Definition
Drug-associated acute kidney injury	Days with exposure to nephrotoxic drugs; exclude (1) patients with ESRD or documented need of dialysis (ICD-9-CM: 585.6x, 585.5x POA or ICD-9-CM: V45.11, V56.xx on current admission) and (2) days after the first pADE day	First day with ≥ 2 times increase in serum creatinine within 5 days after risk day
Drug-associated acute liver injury	Days with exposure to hepatotoxic drugs; exclude (1) patients with use of i.v. acetylcysteine on PTA medication list; or with ICD-9-CM 570, 572.2, 572.4, 573.3 POA, (2) with pADE manifestation on the first day of hospitalization, and (3) days after the first pADE day	First day with ALT increase of at least 5 times ULN within up to 10 days after risk day
Drug-associated acute mental status changes	Days with exposure to opioids, benzodiazepine, barbiturates, anticholinergics, tricyclic antidepressants, or other risk drugs; exclude (1) patients with dementia (ICD-9-CM POA: 290.x, 291.x, 292.82, 294.10, 294.11, 294.20, or 294.21) and (2) with positive substance toxicology screen, or on continuous infusion of sedatives	Days with any of the following: (1) change from negative to positive of CAM-ICU score, (2) change in orientation level from "oriented" and "not disoriented" to "disoriented," (3) change in level of consciousness from "alert" to "irritable," (4) change in RASS score from ≤ 1 to ≥ 2 , (5) change in Riker SAS score from ≤ 4 to ≥ 5 , and (6) change in RSS score from ≥ 2 to 1 AND administration of atypical antipsychotics or haloperidol within 2 hr before and within 6 hr after a mental status change after prior risk day
Drug-associated falls	Days with exposure to drugs that increase risk of falls; exclude days on mechanical ventilation or on continuous infusion of propofol, dexmedetomidine, midazolam, vecuronium, rocuronium, atracurium, cisatracurium, or succinylcholine	Days with falls identified from hospitals' incident reporting system after prior risk day
Drug-associated hypoglycemia	Days with exposure to antidiabetic drugs except α -glucosidase inhibitors	Days with blood glucose <50 mg/dL not followed by another value >80 mg/dL within 10 min after prior risk day
Drug-associated hypotension	Days with exposure to hypotensive drugs	Days with MAP of 60 mm Hg or MAP <70 mm Hg and decrease of at least 40% from the highest value within 24 hr after prior risk day
Drug-associated QT prolongation	Days with exposure to Q-T-interval-prolonging drugs; exclude days after the first pADE day	First day with Q-Tc increase >60 msec over baseline defined as lowest measured Q-TcB before the current value but measured during current admission; or a single Q-Tc >500 msec after prior risk day
Hyperkalemia	Include all patients except patients on dialysis within 10 days before or during admission or with severe leukocytosis ($>70 \times 10^9/L$) or thrombocytosis ($>900 \times 10^9/L$)	Days with a serum potassium concentration of ≥ 5.5 meq/L and a subsequent order to treat hyperkalemia within 12 hr; if loop diuretics, must not have any prior administrations during current admission; if β -agonists, the frequency must be "once" or "once now" and must not have any prior administrations during current admission. If insulin, route must be i.v. and frequency must be "once" or "once now," unit must be "units" (not units/hr), and total dose of ≤ 20 units per administration
Hypokalemia	Include all patients except patients on dialysis within 10 days before or during admission	Days with serum potassium concentration of ≤ 3.0 meq/L and subsequent order of potassium preparations within 8 hr; supplemental potassium doses scheduled before the abnormal potassium value are not considered treatment

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Table 2. Definitions of pADEs and Associated Risk Days^a

pADE	Risk Day Definition (Cohort Entry)	pADE Definition
Hyponatremia	Include all patients except patients with sodium value of <130 mmol/L on admission or on dialysis within 10 days before or during admission	Days with serum sodium concentration of <130 mmol/L
Uncontrolled <i>Clostridium difficile</i> infection	Patients with stool culture positive for presence of toxigenic <i>C. difficile</i> or <i>C. difficile</i> toxins. Denote a patient's first abnormal stool (stool assessment contains one or both of "loose" or "watery") as S1. If the first positive <i>C. difficile</i> culture occurred before the first abnormal stool assessment, then treat that day as S1. Exclude days after the first pADE day.	First day with any of following: (1) stool assessment as "watery" anytime between S10 and S21, (2) stool assessment as "loose" between S10-S21 and with fever or elevated WBCs from baseline within 24 hr before or after (where baseline is the highest WBC count on the day of a positive stool test, the previous day, or the following day), (3) colectomy between days of positive stool culture and S21, or (4) death
Uncontrolled hyperglycemia	Include all patients	Days with at least 2 blood glucose levels of ≥ 250 mg/dL at least 6 hr but less than 24 hr apart (second value counts as time of onset); or 1 single blood glucose level of > 300 mg/dL
Uncontrolled hypertension	Include all patients	Days with SBP ≥ 180 mm Hg or DBP ≥ 110 mm Hg at least 4 hr but less than 24 hr apart (second value counts as time of onset)
Uncontrolled pneumonia	Principal or secondary diagnosis for pneumonia (ICD-9-CM 480-486) with POA flag and chest radiograph within 12 hr of hospital admission	Fourth hospital day with (1) at least 1 respiratory sign or symptom listed below and (2) at least 1 other sign or symptom listed below: Respiratory signs and symptoms (RR ≥ 30 breaths/min, SaO_2 $< 90\%$, PaO_2 < 60 mm Hg, $\text{PaO}_2/\text{FiO}_2$ ratio ≤ 250 , on mechanical ventilation) Other signs and symptoms (temperature > 37.8 °C or < 35 °C, HR > 100 beats/min, leukocytosis [WBCs $\geq 12,000$ cells/mm ³])
Uncontrolled postsurgical pain	First 5 days postsurgery starting at patient's exit time from operating room	Days with at least 2 pain scores of > 7 more than 4 hr apart
VTE	Include all patients except (1) patients with principal or secondary diagnosis of VTE POA: 415.11, 415.13, 415.19, 451.1x, 451.2, 451.9, 453.2, 453.4x, 453.8x, 453.9, 671.4x, 671.5x, 671.94, and 673.2x and (2) days after the first pADE day	ICD-9-CM for VTE not POA and first day with presence of any of the following within 30 days of admission: (1) administration of anticoagulant at therapeutic dose, (2) ultrasound test, (3) V/Q scan, and (4) first chest CT scan (for pulmonary embolism)

^apADE = preventable adverse drug event, ESRD = end-stage renal disease, ICD-9-CM = International Classification of Diseases, Ninth Edition, Clinical Modification, POA = present on admission, ALT = alanine transaminase, ULN = upper limit of normal, CAM-ICU = Confusion Assessment Method for the ICU, RASS = Richmond Agitation-Sedation Scale, SAS = Sedation-Agitation Scale, RSS = Ramsay Sedation Scale, MAP = mean arterial pressure, WBC = white blood cell, SBP = systolic blood pressure, DBP = diastolic blood pressure, HR = heart rate, RR = respiratory rate, SaO_2 = oxygen saturation, PaO_2 = partial pressure of oxygen in arterial blood, FiO_2 = fraction of inspired oxygen, VTE = venous thromboembolism, V/Q = ventilation/perfusion, CT = computed tomography.

gest reduced priority for pharmacist intervention, these ADEs are associated with increased morbidity and mortality and are frequently preventable through adequate prophylaxis, avoidance of risk medications, and monitoring.³⁹⁻⁴⁶

Fourth, our study suggested novel measures to detect uncontrolled pneumonia and uncontrolled *C. dif-*

ficile infections. Existing definitions of uncontrolled infection are considerably variable and focus on measuring infection severity, which is partly related to preexisting comorbidities or susceptibilities that may not be preventable.⁴⁷⁻⁵⁸ We opted to examine prolonged symptoms that could have been resolved with adequate management of infections during hospitaliza-

tion. The measurement of signs and symptoms included composites of discrete data elements from EHRs including assessments for stool abnormality, pathogen culture results, procedures, laboratory test results, and vital signs, which were then validated via expert chart review.

Our study had several strengths and limitations. First, our use of automated EHR-based algorithms allows for broad and low-cost implementation to monitor pADE frequencies over time within or across institutions. An exception from the use of discrete EHR data fields was the measurement of falls where we had to rely on our institutions' incident-reporting system. While incident reporting is subject to underestimation of true cases, our exploratory analysis revealed that the system captured substantially more cases ($n = 974$, 1.16%) than using diagnosis codes for falls ($n = 148$, 0.18%) from the hospital billing data. Natural

Table 3. Frequency Distribution of pADEs per Admission ($n = 83,787$)^a

No. Concurrent pADEs During Admission	No. (%) Admissions
0	56,563 (67.51)
1	19,162 (22.87)
2	5,922 (7.07)
3	1,610 (1.92)
4	428 (0.51)
5-9	102 (0.12)

^apADE = preventable adverse drug event.

Table 4. Incidence of pADEs^a

pADE	No. Admissions in Risk Population ^b	No. pADE Events	pADE ^c Incidence
Uncontrolled postsurgical pain	19,640	13,484	68.66
Uncontrolled pneumonia ^d	1,530	527	34.44
Drug-associated hypotension	43,630	13,394	30.70
Uncontrolled <i>Clostridium difficile</i> infection ^d	675	176	26.07
Uncontrolled hypertension	83,787	10,880	12.99
Uncontrolled hyperglycemia	83,787	9,819	11.72
Drug-associated hypoglycemia	21,840	1,256	5.75
Hypokalemia	82,149	4,511	5.49
Drug-associated QT interval prolongation ^d	61,898	1,672	2.70
Hyponatremia	79,169	2,101	2.65
Drug-associated acute kidney injury ^d	62,561	1,212	1.94
Hyperkalemia	81,931	857	1.05
Drug-associated acute liver injury ^d	69,762	560	0.80
Drug-associated falls	75,036	466	0.62
Drug-associated acute mental status changes	66,875	262	0.39
Venous thromboembolism ^d	74,283	214	0.29

^apADE = preventable adverse drug event.

^bPopulation with at least 1 risk day.

^cNo. days with pADE per 100 admissions in risk population.

^dOnly 1 pADE (the first episode) was counted, and patients were not allowed to reenter the risk population thereafter.

language processing approaches to detect references to falls in case notes may offer more-sensitive measurement alternatives⁵⁹ but were found impractical for broad implementation within the scope of this study.

Of note, our validation approach of measures definitions relied on expert panel review and empirical analysis with focused chart review as needed. If implementation in other institutions is planned for internal quality-improvement efforts or comparisons across institutions, we recommend formal validation of measures algorithms to ensure measures validity. We base this recommendation on the fact that EHR data elements, documentation, and coding practices are not yet standardized across institutions. For example, while we expect that *ICD-9-CM* codes and laboratory test values are generally applicable across institutions, customized fields such as stool appearance may use highly variable value sets.

Expansion of this work to other institutions would also allow for the development of measures for drug-associated respiratory depression, ileus, and hemorrhage, which we had to omit due to low event rates. In selecting candidate pADEs for pharmacist intervention, we intentionally focused on severe conditions and omitted mild cases with weak contributions to patient morbidity. We included ADEs that have been widely publicized patient safety targets, including VTE, uncontrolled infections, and hypoglycemia,⁶⁰⁻⁶³ but more work is needed to expand our measures to include pADEs that could not be measured from discrete data fields, especially as EHR systems advance in standardization and capabilities in reliable ascertainment of data embedded in narratives such as progress notes.

Second, to focus measures on adverse events that are indeed attributable to drug therapy and commonly preventable, we defined risk days and follow-up days, an approach that substantially differs from typical pADE identification methods. Chart review

and voluntary reporting imply some degree of subjective judgments on whether adverse events were drug related and preventable.^{10,64} Although our measures did not formally assess preventability, we chose to define ADEs that are known to be frequently preventable and excluded ADEs in which pharmacists would not be expected to have a large role in prevention. Confining pADE measurement to a more homogeneous population with risk factors often found to be preventable allowed for a stronger focus on events that are relevant to quality improvement. It is important to note that our use of the term pADE is liberal and does not imply that unpreventable ADEs are entirely omitted in the reported frequency estimates. Likewise, our choice of risk windows relied on published literature and may have overestimated or underestimated time periods during which pADEs were attributable to drug exposure. More epidemiologic work on the etiology of pADEs in hospitals is needed to specify measurement periods and quantify the contribution of risk factors for real-time intervention.

Conclusion

Definitions and incidence estimations for pADEs were developed from EHRs. These pADE measures will be advanced for the creation of prediction models to develop an automated prospective C-score for identifying patients at risk for pADEs to prioritize pharmacist intervention.

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